

SUNSPOKE · DESIGN DECISION RECORD

SunSpoke recommendations accepted

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DOCUMENT CONTROL

Document ID	SSP-DDR-002	Version	0.1
Status	Draft	Date	2026-09-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/sunspoke		
Source	docs/decisions/0002-recommendations-accepted.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-09-25	Amish Chadha	Record Amish's acceptance of all open recommendations and what changed in the repo

0002: Recommendations accepted

- **Date:** 2026-09-25
- **Status:** accepted (items 16, 17 and 18 of SSP-DDR-001); items 14 and 15 remain proposed, awaiting Amish

■ Context

After the TRL 3 session, SSP-DDR-001 left five items open. Three carried a recommendation (items 16, 17 and 18) and two did not (items 14 and 15). On 2026-09-25 Amish wrote: "i accept all your recommendations, go with them across all repos." This record lists what that decides for SunSpoke, what changed in the repo, and what stays open. TRL 4 remains on hold by Amish's instruction, and the project stays at TRL 3.

■ Options considered

The options and recommendations are in SSP-DDR-001 (Table 2), SSP-CAL-001 v0.1 sections 4 and 6, and `docs/REVIEW.md` (session 2026-09-25, TRL 3). They are not repeated here.

■ Decision

Table 1. Newly decided items.

# (SSP-DDR-001)	Item	Decision	What changed in the repo
16	Handlebar power switch in series with the INTERLOCK coding resistor; host adapter powered from legacy discharge or the charge inlet	Decided by Amish, 2026-09-25: go with recommendation (adopt)	R10 in SSP-REQ-001 v0.4 now requires the switch in the INTERLOCK loop; SSP-PRC-001 v0.4 records it as decided. The switch was already costed in BOM item 8 and shown on SSP-DWG-001, so cost and geometry are unchanged
17	Motor temperature input on the controller so an overheat becomes a derate, not a cutout (R3)	Decided by Amish, 2026-09-25: go with recommendation (controller with a thermistor input; motor with a winding thermistor)	R3 restated in SSP-REQ-001 v0.4 (derate from 110 °C, winding at or below 120 °C, no abrupt cutout). BOM item 1 \$70 to \$72 (built-in 10 kΩ NTC thermistor), item 3 \$22 to \$25 (thermistor input and derate); bike kit \$183 to \$188. SSP-CAL-001 v0.2 adds a derate calculation. SSP-DWG-001 notes updated, Rev P1 to P2
18	Raise with the SwapCell project that a pack in legacy discharge (state 5) should move to heartbeat discharge (mode 2) without opening the output	Decided by Amish, 2026-09-25: go with recommendation (raise with SwapCell)	Listed under cross-repo actions in <code>docs/REVIEW.md</code> . The interface is governed in the swapcell repo and is not changed here

Effect on requirements and budget

- `budget_usd` stays at \$250. The budget decision (SSP-DDR-001 item 2) already redefined R12 to cover the bike kit only; the bike kit is now \$188, still inside the target with \$62 margin.

- R3 stays **at risk**. With the derate, the base case runs the 500 m design climb at full assist (derating would start only after about 697 m). In the hot-motor case derating starts after about 225 m, and on a climb that does not end the loaded bike slows to about 3.7 km/h instead of stalling. A named motor's thermal data are still needed.
- No requirement is shown to be not met.

Items still open

Table 2. Proposed, awaiting Amish (no recommendation to accept).

# (SSP-DDR-001)	Item	Status
14	First co-design partner and region	Proposed, awaiting Amish. No recommendation; partners are chosen per area later
15	Whether light filing of the donor fork slots (about 0.24 mm per side) is acceptable with torque arms fitted (R4)	Proposed, awaiting Amish. No recommendation until the donor survey

■ Consequences

- SSP-REQ-001 and SSP-PRC-001 move to v0.4, SSP-CAL-001 to v0.2, SSP-DDR-001 to v0.2, and SSP-DWG-001 to Rev P2.
- Specifying and testing the derate on real hardware (thermistor calibration, derate curve, hill test) is TRL 4 work and is on hold by Amish's instruction. Nothing past TRL 3 was started.